Embedded Systems International

Team Embedded Code Review Worksheet (PD4)

Team Identifier: S4-B

Team Name (optional):

Date: Sunday, 28 April 2024

Team Members Participating in the Code Review: Prakarsha Poudel, Marina Mannan, Spencer Opitz, and Preston Gaskill

Submit your document as a PDF file in Canvas under the corresponding project assignment.

Instructions for this assignment:

There are two code review checklists provided (files #3 and #4). As a team, select one of the checklist files for this assignment.

A checklist includes lists of rules that embedded code should adhere to. A particular checklist groups the rules into subsets of rules. A subset of rules is sometimes referred to as a perspective. For this assignment, the subsets are shown in the checklist in each file (file #3 has subsets labeled Reviewer #, and file #4 has subsets labeled Function, Style, etc.).

Each student on a team should select one of the subsets in the team's checklist (e.g., a student might select the Reviewer #1 subset or the Style subset). Then a student picks 2-3 rules listed in their subset. This keeps the workload reasonable for this mock review. Students on the same team may select the same subset if there are more students than subsets in a checklist. However, it would be useful to pick different rules.

Each student keeps their chosen rules in mind when reviewing the team's code (whatever lines of code were selected). A student should write down any issues found in the code in relation to the rules they are responsible for. You don't need to improve the code. You are simply reviewing code with particular rules in mind and identifying potential issues.

Note: If you are unsure about what a rule means, write down what you think it means and apply the rule as you defined it.

Select one of these checklists for the code review:

✓ Peer Review Checklist: Embedded C Code (P. Koopman, ECE CMU, 2018)

____ Embedded System Code Review Checklist (G. Khattak & P. Koopman, July 2012)

These checklists are items 3 and 4 in the Code Review resources for this assignment:

- 1-Better Embedded System SW-Embedded System Code Review Checklist.pdf
- 2-PeerReview-selected-slides-CMU.pdf
- 3-peer_review_checklist-CMU.pdf
- 4-code_review_checklist_v1_010.pdf

1. Select 30-40 lines of code developed for your project.

```
while(1){

    //char command = cyBot_getByte();
    oi_update(sensor_data);

    if (sensor_data->cliffFrontLeftSignal < 20 ||  sensor_data->cliffFrontRightSignal < 20 ||  sensor_data->cliffLeftSignal < 20 || sensor_data->cliffRightSignal < 20) {
        uart_sendStr("\n\r: Hole in the way!");
        oi_setWheels(0, 0);

        //If left hole, turn right
        if (sensor_data->cliffLeftSignal < sensor_data->cliffRightSignal) {
            move_backwards(sensor_data);
            turn_left(sensor_data, 90);
            uart_sendStr("\n\rHit the boundary: turned LEFT!");
        }

        //Else, turn left
        else {
            move_backwards(sensor_data);
            turn_right(sensor_data, 90);
        }
    }
    //If out of bounds, move backwards and stay in the course
    if (sensor_data->cliffFrontLeftSignal > 2700 || sensor_data->cliffFrontRightSignal > 2600) {
        uart_sendStr("\n\r: Hit the boundary!");

        if (sensor_data->cliffLeftSignal < sensor_data->cliffRightSignal) {
            move_backwards(sensor_data);
            turn_left(sensor_data, 90);
            uart_sendStr("\n\rHit the boundary: turned LEFT!");
        }
        else {
            move_backwards(sensor_data);
            turn_right(sensor_data, 90);
            uart_sendStr("\n\rHit the boundary: turned RIGHT!");
        }
    }

    //If left bump, move around it
    if (sensor_data->bumpLeft) {
        uart_sendStr("\n\r: Bumped into object on the left side!");
        move_backwards(sensor_data);
        oi_update(sensor_data);
        turn_right(sensor_data, 80);
        uart_sendStr("\n\r, so I turned RIGHT!");
    }

    //If right bump, move around it
    else if (sensor_data->bumpRight) {
        uart_sendStr("\n\r: Bumped into object on the right side!");
        move_backwards(sensor_data);
        oi_update(sensor_data);
        turn_left(sensor_data, 80);
        uart_sendStr("\n\r, so I turned LEFT!");
```

2. From your selected checklist, identify *perspectives* to be taken by team members. Refer to items 1 and 2 in the Code Review resources to learn about perspectives.

    Too many global variables (control flow)

    Overuse of nested if-else statements (style issue)

    Use of “magic” numbers

3. From your selected checklist, identify several specific rules to be applied during the code review. Each team member should **choose two or three rules** and focus on those given the limited time available for a short mock review session.

    #1: Commenting: top of file, start of function, code that needs an explanation and Proper modularity, module size, use of .h files and #includes

#2: Conditionals should be minimally nested (generally only one or two deep) and Loop entry and exit conditions correct; minimum continue/break complexity
#3: fix concurrency issues (locking, volatile keyword, minimize blocking time) and proper Input parameter checking is done (style, completeness)

4. Record any issues found related to the specific rules considered.
   Nested if-else statements
   Some functions with unclear values passed in (magic numbers)
   Use of global variables